System Vulnerability Assessment

Course code: CYS 330

Course Lecturer: Paul M. Mutunda

Hardware Trojans (HT)

- A **hardware trojan** is a malicious modification of an integrated circuit's circuitry, designed to disrupt operations, steal data, or cause physical harm or death.
- These modifications can be inserted during the design or manufacturing stages, often through untrust third-party components or intellectual property cores.
- They are characterized by their physical representation, activation phase, and action phase, and they can be triggered internally or externally.
- **Detection** is a problem because they are considered to be stealthy (evade detection) nature, requiring advanced methods such as analog detection and power trace analysis, especially in high security environments

Characteristics of Hardware Trojans

1. **Size:** This refers to the number of components added or modified. Trojans can range from small transistors to large circuits.
2. **Distribution:**

(a) **Tight Distribution:** Trojan components are located close together in the layout.

(b) **Loose Distribution:** Components are dispersed across the chip making detection harder

3. **Structure:** Changes in the physical layout can alter the chips (IC) dimensions or component placement

4. **Type**

(a) **Functional trojans:** These involve adding or deleting components such as transistors or logic gates to modify chip's functionality

(b) **Parametric Trojans:** These involve modifying existing components such as thinning wires or weakening transistors, to affect the chip's reliability or performance.

Impact of Hardware Trojans

1. **Data breaches and intellectual property theft:** They can facilitate unauthorised access to sensitive data, leading to breaches and theft of intellectual property.
2. **System Malfunction:** HT can cause critical systems to malfunction, potentially leading to catastrophic failures in safety – critical applications
3. **Performance degradation:** They can degrade system performance over time affecting reliability and efficiency
4. **Reputational damage and financial costs:** Discovery of a hardware trojan can lead to significant reputational damage and financial loses due to recalls and liability issues.
5. **Denial – of – service (DoS) attacks:** HT can be designed to implement DoS attacks, rendering system functionalities unavailable.

Mitigation and Remediation techniques of HT

1. **Detection Techniques** such as Logic testing and side channel analysis could be used. Logic testing is effective for detecting small trojans and affect logic functionality . Side-channel analysis means analysing power consumption or timing to help identify trojans in the hardware.
2. **Collaborative methods:** Using techniques such as flit integrity and dynamic flit permutation in networks-on-chip (NoC) architectures to mitigate trojans
3. Implementing **Message Authentication Codes (MACs)** to ensure data integrity and prevent unauthorised access.

Other attacks targeting the OS/CPU

1. **Theft of service attacks:** These attacks involves manipulating the scheduler to obtain more CPU resources than allocated, often, by exploiting the granularity of timekeeping or the predictability of scheduling tricks .

Mitigation: To mitigate such vulnerabilities, modifications such as using high precision clocks or randomised scheduling can be implemented to prevent attackers timing and resource allocation.

2. Timing attacks: Scheduling algorithms can be vulnerable to timing attacks if they rely on predictable timing mechanisms. This can allow the attackers to manipulate the systems timing to gain unauthorised access or disrupt operations

3. **Denial-of-Service (DoS) attacks:** as discussed earlier.

Other attacks targeting the OS/CPU

4. **Scheduler vulnerabilities in cloud computing (Xen Hypervisor):** In virtualization environments such as Xen, flaws in the scheduler can allow malicious virtual machines (VMs) to dominate the CPU **resources (Memory).** For example, an attacker can manipulate the Xen Credit scheduler to consume up to 98% of the CPU time, depriving other VMs of their fair share. This vulnerability also facilitates theft-of-service attacks in the cloud platforms such as Amazon EC2

5. **Execution unit scheduler contention on AMD CPUs:** It is a side channel vulnerability which affects AMD processors using simultaneous multithreading (SMT) in microarchitectures such as Zen 1 to Zen 4. Attackers can measure contention levels on scheduler queues to potentially leak sensitive information disclosure risks.

Mitigation: Robust scheduler designs and mitigations such as limiting resource monopolization and improving fairness mechanisms.